# Daniel Saromo-Mori

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## Education

**Czech Technical University in Prague** Prague, Czechia

*M.Sc. Cybernetics and Robotics, AI Specialization — Currently Enrolled* *Sep 2024 – Present*

- Awarded merit scholarship for excellent study results in the Winter Semester of Academic Year 2024/2025
- Ranked 1st in Czechia in QS Engineering Ranking 2025

**Politecnico di Milano** Milan, Italy

*M.Sc. Automation and Control Engineering — Weighted Mark: 25.6/30* *Sep 2022 – Feb 2024*

- Completed coursework at institution ranked 13th in the world (QS Engineering 2022)
- Secured 1st place at Pitch Competition 2023 organized by Entrepreneurship Club POLIMI

**Pontificia Universidad Católica del Perú (PUCP)** Lima, Peru

*B.Sc. Mechatronics Engineering & Professional Degree* *Mar 2014 – Nov 2020*

- Graduated in the top 6.66% of the Faculty of Science and Engineering (6th of 32 mechatronics graduates)
- Thesis: "Intelligent spider robot for detecting anti-personnel metallic landmines in uneven terrain"
- Earned Best Bachelor's Thesis award and Innovation Recognition Award at IMECE 2019 (world finals)
- Received Extraordinary Support Funding for Undergraduate Research Thesis; thesis rated outstanding unanimously

## Scientific Publications — 200+ Citations (Google Scholar)

**[C4]** Valdenegro-Toro, M. and **Saromo, D.** "A Deeper Look into Aleatoric and Epistemic Uncertainty Disentanglement," *LXCV Workshop at CVPR 2022*. New Orleans, USA. **200 citations.** Selected for oral presentation.

**[C3]** Bravo, L., **Saromo, D.**, and Villota, E. "Smart Insole Sensor for vGRF Measurement," *9th International Symposium on Sensor Science*. Warsaw, Poland. 2022.

**[C2] Saromo, D.**, Bravo, L., and Villota, E. "Smart Sensor Calibration with Auto-Rotating Perceptrons," *LXAI Workshop at ICML 2020*. Vienna, Austria. Selected for oral presentation.

**[C1] Saromo, D.**, Villota, E., and Villanueva, E. "Auto-Rotating Perceptrons," *LXAI Workshop at NeurIPS 2019*. Vancouver, Canada. Selected for oral presentation.

**[T1] Saromo, D.** "Intelligent spider robot for detecting anti-personnel metallic landmines in uneven terrain," *Pontificia Universidad Católica del Perú*. Lima, Peru. 2020.

**[Under Review]** "Auto-Rotating Neural Networks: An Alternative Approach for Preventing Vanishing Gradients," *Transactions on Machine Learning Research (TMLR)*.

## Professional & Research Experience

**Siemens** Prague, Czechia

*Application Engineer, IT/OT Integration (Working Student)* *Aug 2025 – Present*

- Deliver IT/OT integration solutions at Siemens while concurrently pursuing M.Sc. in Cybernetics and Robotics at CTU Prague

**German Research Center for Artificial Intelligence (DFKI)** Bremen, Germany (Remote)

*Guest Researcher* *Aug 2020 – Dec 2020*

- Extended the Auto-Rotating Perceptron (ARP) concept into a new neural network family (ARNN), implementing dense, recurrent, LSTM, GRU, and convolutional auto-rotating layers
- Validated ARNN performance against equivalent baseline models using the center's GPU clusters, presenting results at the Online Asian Machine Learning School (OAMLS) at ACML 2021
- Collaborated with research advisor Dr. Matias Valdenegro-Toro on architecture design and experimental methodology

**PUCP Applied Robotics and Biomechanics Research Group (GIRAB)** Lima, Peru
*Research Assistant* *Mar 2020 – Jul 2020*

- Applied Auto-Rotating Perceptrons to wearable force sensor calibration, achieving 15x improvement in neural network performance (lower loss) compared to standard architectures
- Co-authored peer-reviewed paper presented at ICML 2020 under the guidance of research advisor Dr. Elizabeth Villota

## Teaching Experience

**PUCP Continuing Education Department** Lima, Peru
*Lecturer — Specialization Diplomas* *Sep 2019 – Present*

- Instructed AI for Games across 12 consecutive semesters (2019-2 through 2025-1), training 300+ professionals in applied AI techniques
- Taught Data Analysis Methods for Time Series in the Diploma in Data Analytics program (2022-1)

**PUCP Center for Advanced Manufacturing Technologies (CETAM)** Lima, Peru
*Lecturer* *Sep 2020 – Sep 2023*

- Delivered ML for Industry across 7 semesters and Python for Data Science across 5 semesters, equipping manufacturing professionals with applied AI and data science capabilities

**National Meteorological and Hydrological Services (SENAMHI)** Lima, Peru
*Lecturer — Peruvian Government Entity* *May 2022 – Jun 2022*

- Designed and delivered introductory AI and ML curriculum for government meteorological professionals

**PUCP Undergraduate School** Lima, Peru
*Teaching Assistant — Faculty of Science and Engineering* *Mar 2019 – Dec 2019*

- Supported instruction in AI (2019-1), Machine Learning (2019-2), and Computer Science Applications (2019-2)

## Honors & Awards

- **Merit Scholarship**, Czech Technical University in Prague — excellent study results (2024)
- **1st Place**, Pitch Competition 2023, Entrepreneurship Club POLIMI, Milan, Italy (2023)
- **LXAI Travel Grant** ($1,860), NeurIPS 2019 — oral and poster presenter, Vancouver, Canada (2019)
- **Innovation Recognition Award** ($250), Old Guard 63rd Annual Oral Competition, World Finals at IMECE, Utah, USA (2019)
- **ASME Travel Award** ($1,500), representing PUCP and South America at ASME IMECE, Utah, USA (2019)
- **1st Place + Technical Award** ($850), Old Guard Oral Presentation Competition, ASME E-FEST South America, Lima (2019)

## Talks & Presentations — 6 Countries

- **Oct 2024**: Auto-Rotating Neural Networks pitch, Dny AI series, *Prague, Czechia*
- **Jun 2022**: Uncertainty Disentanglement paper, LXCV Workshop at CVPR, *New Orleans, USA*
- **Nov 2021**: ARNN poster, Online Asian Machine Learning School at ACML, *Singapore*
- **Jul 2020**: ARP sensor calibration paper, LXAI Workshop at ICML, *Vienna, Austria*
- **Dec 2019**: Auto-Rotating Perceptrons paper, LXAI Workshop at NeurIPS, *Vancouver, Canada*

## Skills & Additional

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| **AI/ML:** | PyTorch, TensorFlow, Keras, Scikit-Learn, OpenCV, OpenAI Gym; Deep Learning (MLP, CNN, ARNN/ARP, DRL), Computer Vision, Transfer Learning |
| **Robotics:** | ROS-1, Jetson Nano, Embedded Systems (Arduino, Raspberry Pi); TIA Portal (PLC/HMI), LabView |
| **Programming:** | Python, MicroPython, C, C++, MATLAB, VBA, LaTeX; NumPy, Pandas, Matplotlib, Plotly, Git, Linux |
| **CAD/CAE:** | SolidWorks, Inventor, AutoCAD, EagleCAD, Altium Designer |
| **Languages:** | Spanish (native), English (proficient), Italian (advanced), French (elementary) |